

N4DSC08 MODBUS RTU Command

MODBUS commands (Function Code, 06/16 control, 03/04 read status)

Notice:

- 1 Data must be HEX format
- 2 The Slave address (Device address) must be the same as the one set, or you can use this command to query the current device address: FF 03 00 FD 00 01 00 24;
- 3 Consistent baud rate and parity
- 4 If communication fails, please short the RES jumper for 5 seconds to restore the factory settings.

1、Instruction introduction

Function code:

Function code	MODBUS address (PLC)	Register address	Remark
04:	30001	0x0000-0x0001 (0-7)	DS18B20 temperature values Unit: 0.1°C
03	40001	0x0080-0x00FF (128-255)	Read special function register (baud rate, 485 address, etc.)
06	40001	0x0080-0x00FF (128-255)	Write a single special function register (baud rate, 485 address, etc.)
16(0x10)	40001	0x0080-0x00FF (128-255)	Write multiple special function registers (baud rate, 485 address, etc.)

All states are mapped to the 4xxx interval register. The user can read or modify the value of the 4xxx interval register to monitor the module input and output status (03 06 16 function code).

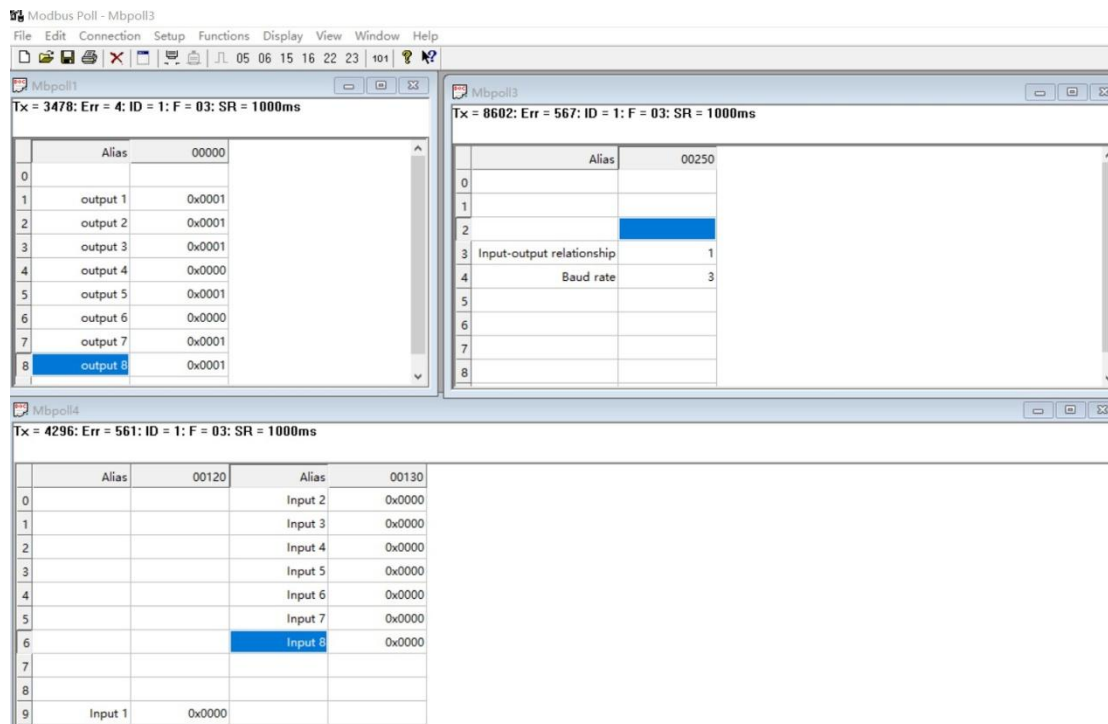
Address	Parameter name	Setting range	Description of parameters	Read /write
0x00A0-0x00A7 (160-167)	DS18B20 Temperature values		DS18B20 temperature values Unit: 0.1°C In case of sensor anomaly. Data is 0X8000 (32768)	R

0x00C0-0x00C7 (192-199)	DS18B20 temperature correction values. If the collected temperature error exceeds 1%, please correct it here.	2 bytes, Input positive temperature value increases; The input negative temperature value decreases	R/W
Special Function Register			
0X00F3 (243)	Sensor probe mode: 0: Mode 0 (default); 1: Mode 1. Note: For sensors with shorter wire lengths, both modes can be used. However, for sensors with wires longer than 5 meters, the default mode may not be able to read data correctly. You can switch to mode 1. If you still cannot read data correctly after switching modes, you can reduce the pull-up resistor, such as changing to 3.9K, 2.7K pull-up resistor.		R/W
0X00F4 (244)	Temperature abnormality judgment value, this register is used to improve temperature abnormality mutation (default value 10) 0: Do not judge abnormality 1: Judge abnormality if the temperature exceeds 0.1 degrees Celsius within 100MS 2: Judge abnormality if the temperature exceeds 0.2 degrees Celsius within 100MS ... 255: Judge abnormality if the temperature exceeds 22.5 degrees Celsius within 100MS		R/W
0X00F5 (245)	DS18B20 Automatic reporting of temperature values Automatic reporting of register (160-167)	0 : Enquiry (default) 1-255 : Automatic reporting of all channel temperature values in seconds 1 : Reported every 1 second 2: Report every 2 seconds 10: Report every 10 seconds Maximum interval 255 seconds	R/W
0x00FB (251)	Restore factory settings	00	R/W
0x00FC (252)	Data return delay	0-1000	R/W

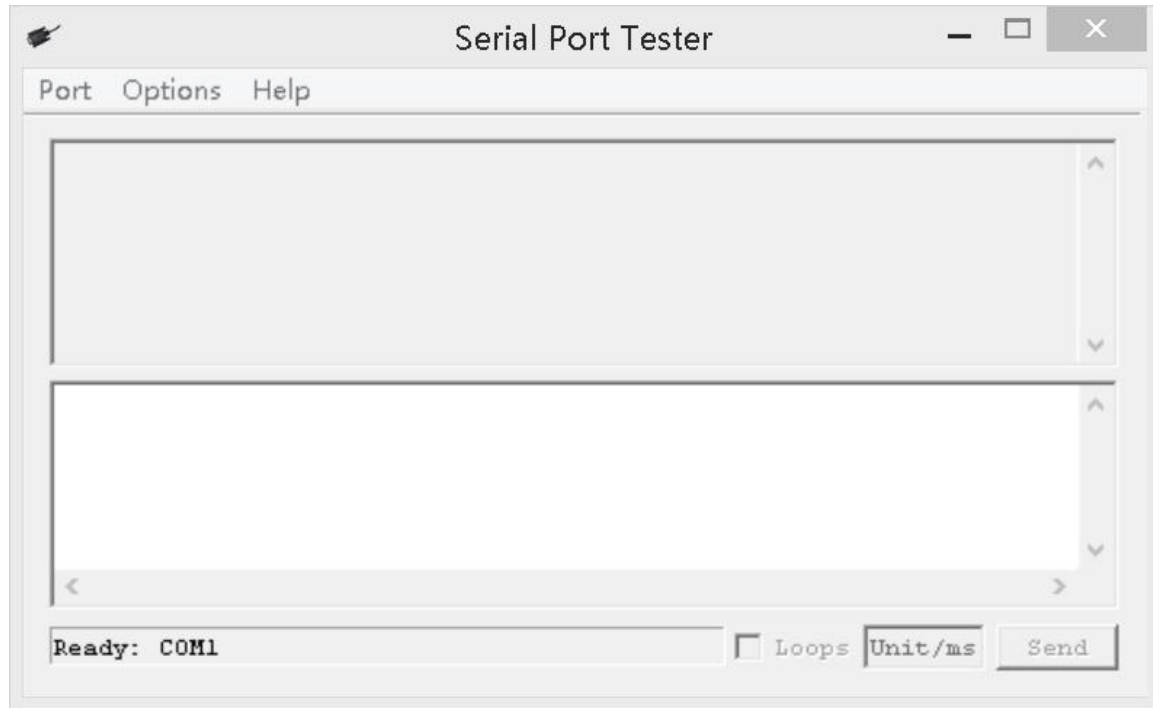
			MS) Set values:0-1000	
0x00FD (253)	RS485 Address / Slave Address	1-247	FF 03 00 FD 00 01 00 24 Query current device address You can also use the command: FF 06 00 FD 00 02 8C 25 to set the device address to 0x02.	R/W
0x00FE (254)	Baud rate	0-255	0:1200 1:2400 2:4800 3:9600 (default) 4:19200 5: 38400 6: 57600 7: 115200 Other: Restore factory settings	R/W
0x00FF (255)	parity bit	0-2	0 :None(default) 1 :Even Parity 2 :Odd Parity	R/W

Default baud rate 9600, 8 data bits, 1 stop bit, no parity bits.

MODBUS commands can be entered using "Modbus Poll" as shown below



It can also be entered using the serial HyperTerminal, as follows



1. Reading DS18B20 temperature values:

Sending frames:

Address Code (1)	Function Code (1)	Register Address (2)	Number of reads (2)	CRC16 (2)
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Return frames:

Address Code (1)	Function Code (1)	Number of bytes (1)	Register value (n)	CRC16 (2)
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PLC Register Address: 30001

Address Code: 0x01~0xF7

Function Code: 0x04

Register Address: 0x0000

Number of readings: 0x0001

Example, reading DS18B20 temperature value:

Sending frames (address 1): 01 04 00 00 00 01 31 CA

Return frames: 01 04 02 00 8F F8 94

01 address code, 04 function code, 02 length, 00 8F means that the current SHT30 temperature reading is 143, which is converted to a temperature value of $143 \times 0.1 = 14.3^{\circ}\text{C}$. F8 94 is a CRC16 check.

In addition the DS18B20 temperature values are also mapped to the 40000 interval register. The DS18B20 temperature values can be read by the user via the 03 function code.

PLC Register Address: 40161

Address Code: 0x01~0xF7

Function Code: 0x03

Register Address: 0x00A0

Number of readings: 0x0001

Example, reading DS18B20 temperature value:

Sending frames (address 1): 01 03 00 A0 00 01 84 28

Return frames: 01 03 02 00 87 F8 26

01 address code, 03 function code, 02 length, 00 87 means that the current DS18B20/NTC/AHT20 temperature reading is 135, which is converted to a temperature value of $135 \times 0.1 = 13.5^{\circ} \text{C}$. F8 26 is a CRC16 check.

Special function registers

1. Set DS18B20 temperature values to be reported automatically:

Sending frames:

Address Code(1)	Function Code(1)	Register Address(2)	Setting values (2)	CRC16 (2)
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Return frames:

Address Code(1)	Function Code(1)	Register Address(2)	Setting values (2)	CRC16 (2)
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PLC Register Address: 40246

Address Code: 0x01~0xF7

Function Code: Write 0x06/0x10; Read 0x03

Register Address: 0x00F5(245)

Setting content: 2 bytes (values 0~255)

Example 1: The current query function should be changed to automatic reporting:

Automatically report in 1 second:01 06 00 F5 00 01 58 38

Automatically report in 5 second:01 06 00 F5 00 02 18 39

Automatically report in 3 second:01 06 00 F5 00 03 D9 F9

Automatically report in 4 second:01 06 00 F5 00 04 38 3B

Automatically report in 5 second:01 06 00 F5 00 05 59 FB

Automatically report in 10 second:01 06 00 F5 00 0A 19 FF

Disable reporting function: 01 06 00 F5 00 00 99 F8

Example: 2: Read the time when the current DS18B20/NTC/AHT20 humidity is automatically reported:

Sending frames: 01 03 00 F5 00 01 94 38

Return frames: 01 03 02 00 01 79 84

01 address code, 03 function code, 02 length, 00 01 is the current DS18B20/NTC/AHT20 Temperaturevalue auto-upload time, converted to 1 in decimal, indicating that the Temperaturevalue is automatically uploaded every 1 second, 79 84 crc16 check.

2. Set DS18B20 humidity values to be reported automatically:

Sending frames:

Address Code(1)	Function Code(1)	Register Address(2)	Setting values (2)	CRC16 (2)
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Return frames:

Address Code(1)	Function Code(1)	Register Address(2)	Setting values (2)	CRC16 (2)
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PLC Register Address: 40247

Address Code: 0x01~0xF7

Function Code: Write 0x06/0x10; Read 0x03

Register Address: 0x00F6(246)

Setting content: 2 bytes (values 0~255)

Example 1: The current query function should be changed to automatic reporting:

Automatically report in 1 second: 01 06 00 F6 00 01 A8 38

Automatically report in 2 second: 01 06 00 F6 00 02 E8 39

Automatically report in 3 second: 01 06 00 F6 00 03 29 F9

Automatically report in 4 second: 01 06 00 F6 00 04 68 3B

Automatically report in 5 second: 01 06 00 F6 00 05 A9 FB

Automatically report in 10 second: 01 06 00 F6 00 0A E9 FF

Disable reporting function: 01 06 00 F6 00 00 69 F8

Example: 2: Read the time when the current AHT20 humidity is automatically reported:

Sending frames: 01 03 00 F6 00 01 64 38

Return frames: 01 03 02 00 01 79 84

01 address code, 03 function code, 02 length, 00 01 is the current AHT20 humidity value auto-upload time, converted to 1 in decimal, indicating that the humidity value is automatically uploaded every 1 second, 79 84 crc16 check.

3. Set 485 address:

Sending frames:

Address Code(1)	Function Code(1)	Register Address(2)	Setting values (2)	CRC16 (2)
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Return frames:

Address Code(1)	Function Code(1)	Register Address(2)	Setting values (2)	CRC16 (2)
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PLC Register Address: 40254

Address Code 0x01~0xF7

Function Code: Write 0x06/0x10; Read 0x03

Register Address: 0x00FD (253)

Setting content: 2 bytes (values 1-247)

Example 1: Set the current device address to 0x02

Sending frames (address 1): 01 06 00 FD 00 02 99 FB

Return frames: 01 06 00 FD 00 02 99 FB

Sending frames (unknown address): FF 06 00 FD 00 02 8C 25

Return frames: FF 06 00 FD 00 02 8C 25

Example 2: Read the current device address (0x0001), only one RS485 device can be connected:

Sending frames: FF 03 00 FD 00 01 00 24

Return frames: FF 03 02 00 01 50 50

FF address code, 03 function code, 02 length, 01 current module address, 50 50 crc16 check.

4. Setting the serial port baud rate:

Sending frames:

Address Code(1)	Function Code(1)	Register Address(2)	Setting values (2)	CRC16 (2)
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Return frames:

Address Code(1)	Function Code(1)	Register Address(2)	Setting values (2)	CRC16 (2)
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PLC Register Address: 40255

Address Code: 0x01~0xF7

Function Code: Write 0x06/0x10; Read 0x03

Register Address: 0x00FE(254)

Setting content: 2 bytes (values 0-7)

For example 1, to change the baud rate to 4800:

Sending frames (address 1): 01 06 00 FE 00 02 69 FB

Return frames: 01 06 00 FE 00 02 69 FB

Baud Rate Corresponding Numbers: 0:1200 1:2400 2:4800 3:9600
 4:19200 5:38400 6: 57600 7: 115200 8: Restore factory settings
Note: 1 The baud rate will only be updated when the module is re-powered when using this command!

2 The factory settings can be restored when the baud rate corresponds to the number 8
For example: 01 06 00 FE 00 08 E9 FC

Example 2, reading the current baud rate:

Sending frames (address 1): 01 03 00 FE 00 01 E5 FA

Return frames: 01 03 02 00 03 F8 45

01 address code, 03 function code, 02 length, 03 means current baud rate is 9600, F8 45 crc16 check.

Baud Rate Corresponding Numbers: 0:1200 1:2400 2:4800 3:9600
 4:19200 5:38400 6: 57600 7: 115200

5. Set Data return delay:

Sending frames:

Address Code(1)	Function Code(1)	Register Address(2)	Setting values (2)	CRC16 (2)
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Return frames:

Address Code(1)	Function Code(1)	Register Address(2)	Setting values (2)	CRC16 (2)
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PLC Register Address: 40253

Function Code: Write 0x06/0x10; Read 0x03

Register Address: 0x00FC(252)

Setting content: 2 bytes (values 0-25)

For example, Set the Data return delay to 200ms

Sending frames (address 1): 01 06 00 FC 00 05 89 F9

Return frames: 01 06 00 FC 00 05 89 F9

Return to the delay time calculation formula: $X = 05 * 40 = 200\text{MS}$

Note: The maximum setting can be 1000MS, more than 1000MS i.e. setting value greater than 25 will initialize the Data return delay.
i.e. : 01 06 00 FC 00 20 48 22 can make the Data return delay re-initialize to 0

6. Set the parity bit:

Sending frames:

Address Code(1)	Function Code(1)	Register Address(2)	Setting values (2)	CRC16 (2)
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Return frames:

Address Code(1)	Function Code(1)	Register Address(2)	Setting values (2)	CRC16 (2)
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PLC Register Address: 40256

Function Code: Write 0x06/0x10; Read 0x03

Register Address: 0x00FF(255)

Setting content: 2 bytes (values 0-2)

For example, setting the parity to even parity:

Sending frames (address 1): 01 06 00 FF 00 01 78 3A

Return frames: 01 06 00 FF 00 01 78 3A

0: No parity (default) 1: Even parity 2: Odd parity 3: Restore initialization

Note: 1 、 The parity bit will be updated only when the module is re-powered when using this command!

2 、 When the setting is greater than 2, re-powering will restore the default 0, No parity.

7. Restore factory settings:

Sending frames:

Address Code(1)	Function Code(1)	Register Address(2)	Setting values (2)	CRC16 (2)
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Return frames:

Address Code(1)	Function Code(1)	Register Address(2)	Setting values (2)	CRC16 (2)
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PLC Register Address: 40252

Address Code 0x01~0xF7

Function Code: Write 0x06/0x10; Read 0x03

Register Address: 0x00FB(251)

Sending frames: FF 06 00 FB 00 00 ED E5

Return frames: FF 06 00 FB 00 00 ED E5

Restore factory settings: Short the RES jumper for 5 seconds